FORWARD DEPLOYED ENGINEERING

FDE-02

1. Executive Summary

- **AI Workers vs. Chatbots:** AI workers move beyond simple conversation by operating inside virtual machines (VMs), utilizing tools, and producing editable digital artifacts.
- **Governance & Security:** Emphasizes authoritative **Systems of Record (SoR)**, resource management, and strict data privacy safeguards (including avoiding raw credit card inputs).

2. Core Concepts & Definitions

Term / Concept	Definition / Explanation
AI Worker	A digital agent capable of understanding tasks, utilizing tools, operating in VMs, and creating editable artifacts.
Session & VM	A temporary cloud computing environment provisioned for an AI worker to execute tasks securely.
Artifact	A durable, editable final output (documents, presentations, designs) enabling two-way human-AI collaboration.
Skill	Procedural knowledge that defines how an AI worker performs a specific task (instructions, best practices).
Connector / MCP	Mechanisms using Model Context Protocol to connect AI workers to external tools, APIs, and business software.
Plugin	A packaged bundle combining skills and connectors for easy distribution and reuse across marketplaces.
SoR (System of Record)	The authoritative source of truth containing trusted knowledge (KSOR) and data (DSOR) for AI workers.

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3. Structured Body Hierarchy

- **AI Worker Fundamentals**
    - **Moving Beyond Chatbots:** Chatbots tell and answer; AI workers execute real-world digital work, maintain state through environments, and generate modifiable assets.
    - **The 10-80-10 Rule Breakdown:**
        - *10% Define Intent*: Human establishes goals and requirements.
        - *80% Execute Task*: AI worker uses intelligence, VMs, skills, and connectors to complete the work.
        - *10% Verify*: Human inspects, edits, and assumes ultimate responsibility for the result.
- **Execution Environments & Artifacts**
    - **VM Sessions:** Each session provisions a dedicated virtual machine with necessary tools; sessions expire and temporary files are eventually destroyed.
    - **Artifact Collaboration:** Shifts interaction from one-way prompts (Human → AI → Answer) to collaborative editing (Human ↔ AI → Shared Editable Output).
- **Modular Architecture: Skills, Connectors, & Plugins**
    - **Skills:** Contain the procedural instructions and workflows ("how").
    - **Connectors / MCP:** Provide integration with external applications and data sources ("where/with what").
    - **Plugins:** Package skills and connectors together into shareable, reusable packages.
- **Governance, Education, & Security**
    - **Systems of Record (SoR):** Authoritative repositories preventing AI hallucinations by supplying verified knowledge and policies.

- **Z Tutory Integration:** Connects educational tools, books, and tracking systems with AI workers.
- **Security Protocols:** Reviewing API data-use policies, using private deployments for sensitive data, and strictly prohibiting raw financial credentials (credit cards) in conversational interfaces.

4. Review & Active Recall (Q&A)

- **Q: What is the primary difference between a chatbot and an AI worker?**
  - **A:** Chatbots provide read-only text responses, whereas AI workers execute end-to-end tasks inside virtual machines, use tools, and generate editable artifacts.
- **Q: How does the 10-80-10 rule divide labor between humans and AI?**
  - **A:** 10% human intent definition, 80% autonomous AI execution within a VM session, and 10% human verification and final ownership.
- **Q: What distinction is made between Skills and Connectors?**
  - **A:** Skills provide procedural knowledge on *how* to execute a task, while Connectors/MCP provide the interface and access to external software and systems.
- **Q: Why are Systems of Record (SoR) critical for AI workers?**
  - **A:** They act as authoritative sources of truth (KSOR for knowledge/policy and DSOR for data), ensuring AI workers base their work on reliable, verified information rather than scattered data.